

To: Divyanshu

From: Rohit

Re: Follow-up to yesterday's 11 answers — FS score slice, plus two corrections

What's changed since yesterday

Three things. Everything else in yesterday's reply stands.

- NEW: the FS score's daily valuation slice has no point values specced anywhere. Proposal below — needs your input first (see question at top of Section 1).
- CORRECTION: I flagged growth-multiple fragility as having no threshold. Wrong — it is specced. Details in Section 2.
- CORRECTION: I flagged the -5 floor as internally inconsistent. Wrong — I misread two separate rules as one. Details in Section 2.

1. FS score — the daily valuation slice

First: what have you already implemented here?

The spec defines $fs_score = fs_quality_base + \text{daily valuation slice (OEY, PE percentile, EV/rev)}$. $fs_quality_base$ is fully defined as $50 + (BQ_RAW \times 2.5)$. The slice is not — it names its three inputs and never says how many points each contributes. The only number anywhere is in the TELUS example (OEY 6.4%, income strong $\geq 3\%$, +5). Nothing for PE percentile, nothing for EV/rev, nothing for the other three business types.

If you've already built something for this, send me what you used and we'll document that instead — no point replacing working logic. The proposal below assumes you haven't, which is my working assumption.

Why this matters: fs_score determines fs_class , and fs_class caps $conviction_score$. So an undefined slice means the cap is being applied off a number nobody can reconstruct. This is also the most likely explanation for the CRM discrepancy (displayed +1 / CANCEL vs. BQ +8 and tax -5 = +3, which should be REDUCED BUY) — worth checking against whatever you've got before we treat it as a separate bug.

Two design points before the tables:

- Symmetric, not penalty-only. The valuation tax is penalty-only (always ≤ 0). The slice must not be — if all three inputs only ever subtract, fs_score can never exceed $fs_quality_base$, and a cheap high-quality business gets systematically pushed into a lower FS class than it deserves. Cheap should add, expensive should subtract.
- Direction differs by input. PE percentile and EV/rev: low = cheap = positive. OEY runs the other way — it's a yield, so high OEY = cheap = positive. Easy to get backwards.
- Sizing: $\sim \pm 10$ per input, $\sim \pm 30$ total. FS class boundaries are 25 points apart (weak <25, moderate_low 25, moderate 40, moderate_high 55, strong 75). ± 30 gives valuation roughly one class of movement — enough to matter, not enough to erase the BQ half of the score.

PE percentile (own 20Y history) — self-calibrating, so one table for all business types:

PE percentile	Points
Below 20th	+10

20th–40th	+5
40th–60th	0
60th–80th	-5
Above 80th	-10

OEY — anchored to the existing per-type thresholds in Section 5.1, not new numbers:

OEY vs. business-type thresholds	Points
At or above 1.5x the 'strong' threshold	+10
At or above 'strong' threshold	+5
Between 'strong' and 'expensive'	0
At or below 'expensive' threshold	-5
At or below 0.5x 'expensive' threshold	-10

Worked example, saas (strong $\geq 5\%$, expensive $< 1\%$): $\geq 7.5\% = +10$, $\geq 5\% = +5$, $1-5\% = 0$, $\leq 1\% = -5$, $\leq 0.5\% = -10$. Same table auto-adapts to compounder/cyclical ($\geq 4\%$) and income ($\geq 3\%$) — no separate table per type needed.

EV/forward revenue — reuse the existing Section 5.4 tier boundaries per business type:

EV/fwd rev vs. existing tiers	Points
Below Tier 1	+10
At Tier 1	+5
At Tier 2	0
At Tier 3	-5
At Tier 4 or above	-10

For banks, this row uses Price-to-Tangible-Book instead of EV/rev, consistent with the P/TBV vs. ROE approach in the consolidated note. Note this supersedes the simpler P/B tiers I gave in yesterday's Q1 answer — use the consolidated note's method, it's the more rigorous one.

Sanity check on CRM (BQ +8, so $fs_quality_base = 70$): PE 91st pctile = -10, OEY 1.8% = 0, EV/rev 9.2x (saas Tier 3) = -5. Slice total -15, $fs_score\ 55 = moderate_high = no\ cap$. Conviction stays +3 = REDUCED BUY 40%. Sensible: expensive enough to cost it a class, not enough to gate the trade. Worth noting it does NOT produce 'weak' — which points to the displayed +1 being a coding issue rather than the missing spec.

2. Two corrections to yesterday's note

Growth-multiple fragility — I was wrong, it is specced.

Section 5.3 of the consolidated note defines it: EV/fwd rev $\geq 4x$ AND revenue growth $\geq 15\%$, both true, gives -2. Build as written. The rationale there is worth reading if you haven't — fast growth justifies a higher price but doesn't remove the separate risk of that growth decelerating, which is why it's scored separately from growth_trajectory rather than netted against it. Nothing to resolve here; ignore my earlier flag.

Source: Consolidated Note (28 July) — Section 5.3 (the rule itself) and Section 7 (plain restatement, confirms both NVDA and GOOGL clear the $\geq 15\%$ trigger). Also Session Notes (28 July) Section 4.3, same rule and rationale.

The -5 floor — also wrong, I misread two separate rules as one.

The actual rule is simple and universal: any stock above 4x forward revenue gets a valuation tax floored at -5 minimum. I'd been reading that against the per-type 'floor trigger' column in the v5 spec's Section 5.4 table and concluding the numbers contradicted each other. They don't — they're different things. Build the 4x/-5 rule as specced.

One real exception, from Section 5.5 of the consolidated note: `high_margin_hardware` is exempt from the -5 floor, because the floor's rationale (execution AND multiple maintenance must both go right) doesn't hold when the multiple is already backed by current profit rather than a growth story. That bucket also uses EV/fwd EBITDA or margin-normalised EV/Rev instead of raw EV/Rev, and is exempt from the saas-specific modifiers. That wasn't in yesterday's reply and should be.

Source: Consolidated Note (28 July) — Section 1 for the base 4x / -5 rule, Section 5.2 confirming it applies regardless of business_type, and Section 5.5 for the high_margin_hardware exemption. All three were already in your note; I read past them.

3. Still genuinely open

Short list, so nothing gets lost:

- FS slice point values — pending your answer on what's already built.
- Multi-segment tie-break: which `business_type` wins when a company spans two GICS buckets. Consolidated note flags this as needing an explicit dominant-segment-by-revenue rule rather than a silent default.
- CEO tenure and insider ownership sourcing for non-US names — DEF14A and EDGAR/SEDI don't cover the Indian and Korean tickers already live in the universe.
- Yield-trap market thresholds for Korea/Japan/China — deprioritised (none of the current KR/JP/CN names are dividend payers) but needed before any genuine payer from those markets enters.

4. Working mockup — engine_layers_spec.html (attached)

Attached is a clickable HTML page covering the depth-view Engine Layers panels: Business Quality, Valuation Tax, FS Cap and Yield Trap. Worth opening this alongside the tables above — it shows the slice logic rendered end-to-end rather than described, which is quicker to sanity-check than reading the numbers cold.

Same visual system as `ce_developer_spec_latest.html`, so it drops into the existing depth view rather than being a separate design. Four preview tickers along the top let you step through the cases:

- CRM (saas) — the +1 vs +3 discrepancy, with computed vs. displayed `fs_class` shown side by side.
- HDFCBANK.NS (bank) — the `business_type` branching, i.e. which rows swap out for a bank.
- 005930.KS — the short PE-history flag (6 years, not 20) and the undefined Korea yield-trap threshold.
- UNMAPPED-1 — the `coverage_incomplete` empty state.

Two things to know before you read it. First, all figures in it are illustrative — they demonstrate the mechanism, they are not from a live run, so don't treat any number there as audited. Second, it predates the two corrections in Section 2: it still shows growth-multiple fragility and the floor trigger as 'not yet defined'. Both are in fact

specced, per Section 2 above — the tables in this document are correct, the mockup is not, on those two rows only.

The preview-ticker bar at the top is scaffolding for reviewing it and comes out before deploy — the real page reads the selected ticker. Same for the lookup box, which is a stub for a separate feature (look up any ticker on demand, not just ones with a live signal) that we should scope separately.

Rohit